JAN-S-683

30 SEPTEMBER 1948

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

STEEL, COPPER-PLATED, STRIP (FOR SMALL ARMS BULLET JACKETS)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

A. APPLICABLE SPECIFICATIONS

A–1. The following specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

Federal Specification QQ–M–151—Metals; General Specification for Inspection of.

B. TYPE AND CONDITION

B–1. This specification covers one type of copper-plated steel strip for small arms bullet jackets.

B–2. *Condition.*—Copper-plated steel strip shall be furnished in the following conditions, as specified:

Annealed.
Not annealed.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*

C–1a. The copper-plated steel strip shall be a composite material consisting of a steel core with a layer of copper on each side of the steel.

C–1b. Unless otherwise specified, the steel shall be rimmed or semi-killed with aluminum.

C–1c. Sufficient discard shall be removed from the top and bottom of each steel ingot to insure freedom from pipes and harmful segregations.

C–2. *Workmanship.*

C–2a. The plating on the strip shall be continuous and firmly adherent to both sides of the steel core.

C–2b. The steel core and the copper plating both shall be of uniform quality and condition and shall be free from harmful oxidation, scale, chips, acid, dirt, dented or bent edges, pipes, laminations, slivers, laps, cracks, deep scratches, wrinkles, and other injurious defects.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical composition.*

E–1a. *Steel.*—The chemical composition of the steel core, as determined by Government analysis, shall meet the requirements shown in table I.

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TABLE I.—*Chemical requirements of steel core.*

Carbon, percent, max	.13
Manganese, percent	.27–.53
Silicon, percent, max	.07
Phosphorus, percent, max	.05
Sulfur, percent, max	.06

E–1b. *Copper.*—The copper used for plating shall have a purity of not less than 99.90 percent copper.

E–1c. The contractor shall furnish an analysis of the steel used in the core and an analysis of the copper used in the plating, showing the percentages of the elements designated.

E–2. *Hardness of annealed strip.*—When strip is specified annealed, the hardness of the steel core shall not exceed Rockwell B60.

E–3. *Dimensions and tolerances of plated steel strip.*

E–3a. *Thickness.*—The thickness and tolerances on thickness of copper-plated strip shall be as specified in the invitation for bids and in the contract or purchase order.

E–3b. *Width.*—The width and permissible variations therein of copper-plated steel strip shall be as specified in the invitation for bids and in the contract or purchase order, or as agreed upon by the purchaser and the contractor. Unless otherwise specified, the strip shall be furnished with a commercial No. 3 slit (not filed) edge.

E–3c. *Camber.*—The maximum permissible camber of strip shall be as specified in the invitation for bids and in the contract or purchase order, or as agreed upon by the purchaser and the contractor.

E–3d. The thickness of the copper plating shall meet the requirements shown in table II.

TABLE II.—*Thickness of copper plating.*

Outer thickness, minimum	0.0008 inch.
Inner thickness, minimum	.0001 inch.

E–4. *Work test.*—The material shall draw to the required shape without developing any defect which will cause it to be unsuitable for the purpose intended.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Lot.*—Unless otherwise specified, a lot shall consist of plated strip with the steel core from the same heat or melt of steel, of the same nominal thickness and width, and of same condition of hardness, submitted for inspection at one time.

F–2. *Sampling.*—Samples for the chemical, hardness, and thickness of plating tests shall be taken from 2 percent of the cut lengths or coils in each lot, provided that at least 2 cut lengths or coils shall be sampled from any lot. The samples shall be sufficiently flat to use for hardness test specimens. Each sample shall be at least 24 inches long and the full width of the material supplied.

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F–3. *Gaging.*—At least 5 percent of the cut lengths or coils from each lot shall be selected by the inspector and gaged for thickness and width.

F–4. *Government analysis.*—The sample for Government analysis of the steel shall consist of at least 2 ounces of clean mixed drillings taken from the samples selected for test after the removal of the copper plating.

F–5. *Thickness of plating.*—

F–5a. The thickness of the plating shall be determined by one of the following or such other methods as approved by the contracting officer:

(1) Microscopic examination of a suitable cross section.
(2) Stripping the plating from a measured area and determining the loss in weight.
(3) A magnetic method.

F–5b. When the stripping method is used, the following procedure shall be used:

A sample of the plated strip 1 inch by 1 inch shall be weighed and then dissolved in a solution consisting of 500 gm. per liter of chromic acid and 50 gm. per liter of sulfuric acid at a temperature of 180 to 212°F. until a constant weight is obtained. Calculate for thickness of the plating.

F–5c. When a magnetic method is used, the accuracy of the instrument shall be checked against certified plates at least once a day.

F–5d. In case of dispute, the microscopic method shall be used.

F–6. *Hardness test.*—The Rockwell B hardness of the steel core, after removal of the copper plating, shall be determined every 4 inches along a diagonal of the surface of each sample selected for test.

F–7. *Work test.*—At least 1 percent by weight of the material in each lot shall be submitted to a work test comprising all operations used in the manufacture of copper-plated steel bullet jackets.

F–8. *Rejection.*—If a lot fails on one or more of the requirements of this specification, the entire lot shall be rejected.

F–9. *Resubmission.*—Upon the written request of the contractor addressed to and approved by the contracting officer, a lot of rejected material may be resubmitted and retested. Such resubmission, however, is contingent upon the prior elimination by the contractor from such lot of all material failing to meet the specification, or upon the reprocessing of the material in case of failure to comply with hardness and work test requirements.

G. PACKING AND MARKING FOR SHIPMENT

G–1. Material shall be supplied in cut lengths or coils, as specified in the inquiry or invitation for bids and in the contract or purchase order.

G–2. *Packing.*—

G–2a. All materials shall be properly segregated by lot when prepared for shipment.

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G–2b. All material shall be protected against corrosion in transit.

G–2c. Material shall be packed in such a manner as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

G–3. *Marking.*—

G–3a. Each bundle of cut lengths or package of coils shall be suitably marked with the name of the material, the size, lot number, weight, the name of the contractor, steel heat number, and the number of the contract or purchase order. That side of each coil or each cut length having the greater thickness of copper plating shall be identified by stamping or other suitable means.

G–3b. Shipments shall be marked with the name of the material, the size and quantity contained therein, as defined by the contract or purchase order, the name of the contractor and the number of the contract or purchase order.

H. NOTES

H–1. This specification replaces Frankford Arsenal Tentative Specification No. FXS–581.

H–2. Copies of this specification may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

Notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings or specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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